		$= 300$ m
7	**D**	The hollow sphere has smaller mass than the solid sphere, hence has a smaller weight. So, the hollow sphere will attain terminal velocity sooner.

No.	Answer	Solution
		The terminal velocity of the hollow sphere will also be lower than that attained by the solid sphere. Option A is incorrect because the initial portion of the displacement-time graph suggests both the hollow and solid spheres are released from rest, which is the not the case. Option B is incorrect for the same reason as Option A. The hollow and solid spheres are released from a hot-air balloon moving downwards at a constant speed. Option C is incorrect because the displacement-time graph suggests the solid sphere attains terminal velocity before the hollow sphere. Option D is correct because the displacement-time graph suggests the hollow sphere attains terminal velocity before the solid sphere. Both spheres have a constant downward speed just before they are released.
8	B	By Newton's 2nd law, braking force *F* is directly proportional to rate of